

disturbances, which naturally costs it some energy. A familiar example of this phenomenon is the sonic boom behind supersonic (faster-than-sound) planes or the conical wake that ducks leave on the pond's surface. When charge particles move in a medium with speeds faster than the speed of light in that medium, a kind of radiation called the Cerenkov radiation is emitted by them. It has been predicted by Cohen and Glashow (Nobel laureate) that in a similar manner, neutrinos moving faster than light should emit some such radiation and in the process lose out energy in a specified manner. The ICARUS team has reported that they have seen no such effect in the energy distribution of the neutrinos."

He also points out another reason to doubt the 'faster than light' phenomenon. "When the famous supernova explosion happened in 1987, the light and neutrino bursts from it came more or less at the same time (with a few hours' difference), whereas OPERA's claim would imply that the neutrinos should have reached several years earlier. Considering the fact that the time taken for light is only estimated in the OPERA experiment we must be more patient and wait for a more rational explanation of the observation to surface. So it looks that there is no need to doubt the Einstein's theory of relativity at this time."

It is to be noted that OPERA's and ICARUS' efforts were experiments and not discoveries. The difference between the two is humongous. While even the smallest discovery might be accepted without doubt, no experiment would be absolutely acceptable unless verified by a third party. It is believed that some labs in the US and Japan have undertaken to verify this observation.

Prof. Karbelkar adds, "The relativity theory has been verified in so many ways that even if the new OPERA result is confirmed, it may not entirely scrap Einstein's theory; just as Einstein's theory didn't overthrow

Potential Impact

Time dilation and space contraction. In the theory of relativity, time dilation is an observed difference of elapsed time between two events as measured by observers either moving relative to each other or differently situated from gravitational masses. An accurate clock at rest with respect to one observer, may be measured to tick at a different rate when compared to a second observer's own equally accurate clocks.

This effect arises neither from technical aspects of the clocks, nor from the fact that signals need time to propagate, but from the nature of space-time itself. In special relativity (or, hypothetically far from all gravitational mass), clocks that are moving with respect to an inertial system of observation are measured to be running slower (refer Wikipedia).

Similarly, there are other concepts such as space and length contraction. For example, an observer detects a decrease in the length of objects that travel at any non-zero velocity relative to that observer. It is interesting to note that while time expands from the perspective of the stationary observer, space contracts from the perspective of the moving observer.

However, both time dilation and space contraction are not noticeable at everyday speeds. The effect of time dilation has been experimentally confirmed, thanks to very precise caesium clocks that can measure extremely small periods of time. Unfortunately, time dilation is completely outside human experience because we have not yet devised a way of travelling at speeds where relativistic effects become noticeable. Even if you spent your whole life in a jet plane that moves at supersonic speed, you would barely win a second over your contemporaries on the ground.

Not even today's astronauts can perceive the Lorentz contraction. Imagine you are a cosmonaut on board space station Mir which is moving at 7700 metres per second relative to the earth. Looking down upon Europe from space you would see the entire 270-kilometre east to west extent of Switzerland contracted by a mere 0.08 millimetre. If, however, neutrinos really travel faster than light, time dilation and space contraction could assume a new meaning, and possibly become experienceable.

Time travel. Time travel is the concept of moving between different points in time comparable to how one would move between different points in space. So far only hypothetical, time travel would facilitate movement to any point in the past or future without the need for the traveler to experience the intervening period in its normal rate.

So far the relation between cause and effect as explained by Einstein's theory of special relativity provides for the movement of time only in one direction. This is based on the underlying theory that nothing can move faster than light. However, if it is true that neutrinos travel faster than light, it could open the possibility of one travelling back in time.

Conversion of mass into energy and vice versa. When charged particles move in a medium with speeds faster than the speed of light in that medium, they radiate some energy. So if neutrinos travel faster than light in vacuum, there must be some energy that can be tapped from it.

Any immediate impact on electronics and communication? Prof. Karbelkar feels, "It is very difficult to predict the electronics and communication engineering implications of the claimed findings. On the domestic ground (1000km range), a head-start of a mere 60-billionths of a second might not prompt people to switch over to neutrino-based communication. Remember also that the OPERA team at the receiving end (the Gran Sasso lab) uses a really colossal detector (10x10x20 m³ with a mass of 4000 tonnes). Not very suitable for routine communication. Perhaps for inter-world galactic communications, this, if true, may help!"

the previous one entirely but only brought out its limits of validity. But as I have mentioned earlier, it is a bit too early to treat the OPERA result as firmly established."

So we need to wait and see. No hurry to rewrite our textbooks! ●

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